



JAIPUR NATIONAL UNIVERSITY

Semester/Year End / Back Examination Dec.2024

Name of School:	School of Engineering and Technology
Name of Programme / Course:	B.Tech. (Computer Science) /B.Tech. CSE(Hons) in Artificial Intelligence and Machine Learning/B.Tech. CSE (Hons) in Cyber Security
Semester/Year:	III
Paper Code:	BTECSVA301T22
Title of Paper:	Internet of Things
Time: 3 Hours	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

- This question paper consists of two sections, Section A and Section B.
- Section A consists of seven questions and Section B consists of six questions.
- Attempt any five from Section A and any four from Section B.
- Marks allotted are mentioned against each question.

Question Number	Question	Marks Allotted
Section A		
Q.1	Define the Internet of Things (IoT) and discuss its significance in modern society. What are the primary elements of an IoT ecosystem?	4
Q.2	Explain the role of Machine-to-Machine (M2M) and Wireless Sensor Networks (WSN) protocols in IoT ecosystems.	4
Q.3	Explain the role of the Network layer in IoT protocols like Zigbee. What are the key responsibilities of this layer, and how does it facilitate communication between devices in a wireless network?	4

- b) What are the foundational technologies and principles that underpin WoT, and how do they contribute to the integration of web standards with IoT devices? 5
- Q.5 a) What differentiates Smart Applications from traditional applications, and how do they leverage IoT data to deliver enhanced functionality? 6
- b) Discuss real-world examples of Smart Applications and their impact on user experience and decision-making. 4
- Q.6 Compare and contrast the Web of Things (WoT) and the Internet of Things (IoT). What are the key differences in terms of architecture, communication protocols, and data management? 10

Q.4 Define IoT architecture and explain the significance of open-source architectures in the Internet of Things. 4

Q.5 What are the specific security concerns associated with WoT applications, and how can standardization and middleware solutions help mitigate these risks? 4

Q.6 What is the role of security in the Web of Things architecture. What are the specific security concerns associated with WoT applications, and how can standardization and middleware solutions help mitigate these risks? 4

Q.7 Describe the OIC architecture in detail. What are the key components and layers of the OIC architecture? 4

Section B

Q.1 Discuss the various business drivers that motivate organizations to implement IoT solutions. How do IoT technologies lead to cost reduction, increased efficiency, and improved customer experiences? Provide case studies or examples to illustrate your points. 10

Q.2 Examine the security issues associated with IoT protocols. What are the common vulnerabilities in IoT communication protocols, and how can they be mitigated? 10

Q.3 Explain IEEE 802.15.4 standard and its significance in the context of IoT communication protocols. How does this standard support low-rate wireless personal area networks (LR-WPANs)? Discuss the key features and limitations of IEEE 802.15.4, and provide examples of its applications in IoT systems. 10

Q.4 a) Discuss the two pillars of the Web of Things. Provide examples to illustrate the relationship between these pillars. 5